

de Sitter is the Spinless Vacuum. That is Not a Nobel Theorem.

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Abstract

Cosmology in this program is metric. Torsion is algebraic and vanishes wherever the spin density vanishes, so it is not a cosmological limitation.

The metric sector of the New Standard Model, in vacuum, is Einstein’s equation with cosmological constant. If $\Lambda > 0$ and $T_{\mu\nu} = 0$, the unique complete simply-connected maximally symmetric solution is de Sitter space of radius $\ell = \sqrt{3/\Lambda}$. That identity is **[P]**. It is Einstein gravity, not an entanglement selection of our universe.

Faulkner–Guica–Hartman–Myers–Van Raamsdonk cannot be copied onto the cosmological horizon (Paper 17). Jacobson does not become **[P]** for the Standard Model (Paper 18). Those two facts are why “emergent gravity yields de Sitter” is not a prize theorem. The vacuum is de Sitter. The bookkeeping does not yet select it.

1 The vacuum

When the spin density vanishes, the Cartan equation forces torsion to zero and the NSM metric equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \kappa = 8\pi G. \quad (1)$$

Set $T_{\mu\nu} = 0$ and $\Lambda > 0$. The unique (up to isometry) complete simply-connected Lorentzian 4-manifold of constant positive curvature is de Sitter space, with

$$\Lambda = \frac{3}{\ell^2}, \quad R_{\mu\nu} = \Lambda g_{\mu\nu}, \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0. \quad (2)$$

In the flat FLRW chart this is $a(t) = e^{Ht}$ with $H^2 = \Lambda/3$, which saturates both Friedmann equations at vanishing density. **[P]** (Einstein+ Λ ; machine-checked).

Torsion does not appear. A late-universe spin average is an estimate about matter, not a correction to (??).

2 What this is not

It is not a derivation of Λ from entanglement. It is not an FGHMV theorem about a cosmological CFT. It is not Hehl–Datta. It is not a reason to award a Nobel Prize.

A prize-grade claim would be: the entanglement first law, with a *derived* sign, is equivalent to linearized Einstein+ Λ about de Sitter in the universe we inhabit. Paper 17 proves that copying

the AdS argument is false. That gap is the one that still decides the case. This note does not fill it.

Admission. *I wanted de Sitter. The vacuum is de Sitter. I did not make emergent bookkeeping select it. I will not write that it did.*

References

- [1] S.W. Hawking and G.F.R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge (1973), Ch. 5.